SkillSync

DBMS PROJECT REPORT

Project Documentation and Demonstration

Student	SRN
Shamanth SK	PES1UG24AM258
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GitHub Repository: https://github.com/satk4ll/DBMS_MP

1. Introduction

SkillSync is a web-based student project collaboration platform. It helps students discover academic projects, review required skills, apply to participate, form teams, and share project-related notes. The application combines PostgreSQL for structured relational data with MongoDB for document-oriented collaboration notes.

The application is built with Next.js, React, and TypeScript. Its pages and server-side API routes provide the user interface and database access. Student authentication and server-side role checks protect account-specific and administrative operations.

2. Project Objectives and Scope

- Provide a central place to browse student projects and their required skills.
- Maintain student profiles, skills, projects, applications, teams, and memberships.
- Allow students to apply to projects and track applications.
- Support team creation and membership based on accepted applications.
- Provide authenticated project collaboration notes.
- Demonstrate relational and document-oriented database integration.

The current implementation uses MongoDB as its NoSQL database. A vector database, embeddings, and vector similarity search are not implemented in this version.

3. Student Details

Role	Name	SRN
Person 1	Shamanth SK	PES1UG24AM258
Person 2	Satyanarayana Kallakuri	PES1UG24AM257

4. Abstract

SkillSync provides a centralized platform where project owners can publish projects and specify required skills, while students can review projects and apply to participate. The system supports application tracking, team formation, and project-related collaboration notes.

PostgreSQL stores students, skills, projects, applications, and teams using relational constraints and database features. MongoDB stores collaboration notes as documents. Next.js API routes connect the frontend to both databases. Authentication uses bcrypt password hashing and signed HttpOnly session cookies, while server-side role checks restrict administrative actions.

5. Technology Stack

Component	Technology	Purpose
Frontend	Next.js 15, React 19, TypeScript	Pages and user interface
Styling	CSS	Application styling
Relational DB	PostgreSQL	Structured entities and relationships
NoSQL DB	MongoDB	Collaboration-note documents
Database drivers	pg, mongodb	Node.js database connectivity
Authentication	bcryptjs, HMAC-SHA-256 session	Password hashing and signed session
Version control	Git, GitHub	Source control and hosting

The local .env.local file contains private configuration and is excluded from GitHub. A safe .env.example template is included in the repository.

6. System and Database Design

6.1 Application architecture

The browser communicates with the Next.js application. Frontend pages call API routes. These routes validate input, identify the authenticated user, enforce permissions, and communicate with PostgreSQL or MongoDB. PostgreSQL is the system of record for structured entities; MongoDB stores collaboration-note documents.

6.2 Relational tables

Table	Purpose
STUDENT	Student profile details.
SKILL	Skill catalogue.
STUDENT_SKILL	Student-to-skill mapping and proficiency.
PROJECT	Project details, owner, status and creation time.
PROJECT_SKILL	Project-to-required-skill mapping.
APPLICATION	Student applications and their statuses.
TEAM	Teams associated with projects.
TEAM_MEMBER	Team membership and role.
user_account	Login identity, password hash, role and linked student.

6.3 NoSQL data

MongoDB stores collaboration notes as documents. The notes API requires authentication and derives the author identity from the verified session.

7. SQL – Database Commands and Features

7.1 DDL

The SQL schema scripts define tables with CREATE TABLE, primary keys, foreign keys, UNIQUE constraints, CHECK constraints, and NOT NULL constraints. The authentication schema is maintained in database/auth-schema.sql.

```
-- Representative DDL pattern\nCREATE TABLE student (\n  student_id SERIAL PRIMARY KEY,\n  name VARCHAR(100) NOT NULL,\n  email VARCHAR(150) UNIQUE NOT NULL,\n  department VARCHAR(80),\n  year INTEGER,\n  experience_level VARCHAR(30)\n);\n\n-- ALTER and DROP examples are included in queries.sql.
```

7.2 DML and query operations

database/queries.sql demonstrates INSERT, SELECT, UPDATE, DELETE, joins, aggregate functions, GROUP BY, HAVING, ORDER BY, subqueries, EXISTS, and transaction operations using BEGIN, COMMIT, and ROLLBACK.

```
-- Join projects with their owners\nSELECT p.project_id, p.title, s.name AS owner_name\nFROM project p\nJOIN student s ON s.student_id = p.owner_id;\n\n-- Count applications per project\nSELECT p.title, COUNT(a.application_id) AS application_count\nFROM project p\nLEFT JOIN application a ON a.project_id = p.project_id\nGROUP BY p.project_id, p.title\nORDER BY application_count DESC;
```

7.3 Advanced database concepts

- Indexes support frequently used filters and joins.
- The project_summary view summarizes projects, required skills, and applications.
- The set_project_in_progress() trigger function updates a project when an application is accepted.
- A GIN full-text index supports searching project descriptions.
- Transactions are demonstrated through controlled DML tests and rollback.

The demonstration dataset contains 4 students, 6 skills, 11 student-skill mappings, 3 projects, 7 project-skill mappings, 3 applications, 2 teams, and 4 team members. The project_summary view and trigger function were checked. A test transaction was rolled back to preserve the existing data.

8. NoSQL – MongoDB Commands

MongoDB is used for collaboration notes. The application connects through the MongoDB Node.js driver using MONGODB_URI. Authenticated users can retrieve and create notes. The notes API uses the signed-in user's identity rather than trusting an author value supplied by the client.

```
// Open MongoDB shell\nmongosh\n\n// Select application database\nuse skillsync\n\n// List collections\nshow collections\n\n// Inspect note documents (if present)\ndb.notes.find().pretty()\n\n// Count note documents\ndb.notes.countDocuments()\n\n// Check server availability\ndb.runCommand({ ping: 1 })
```

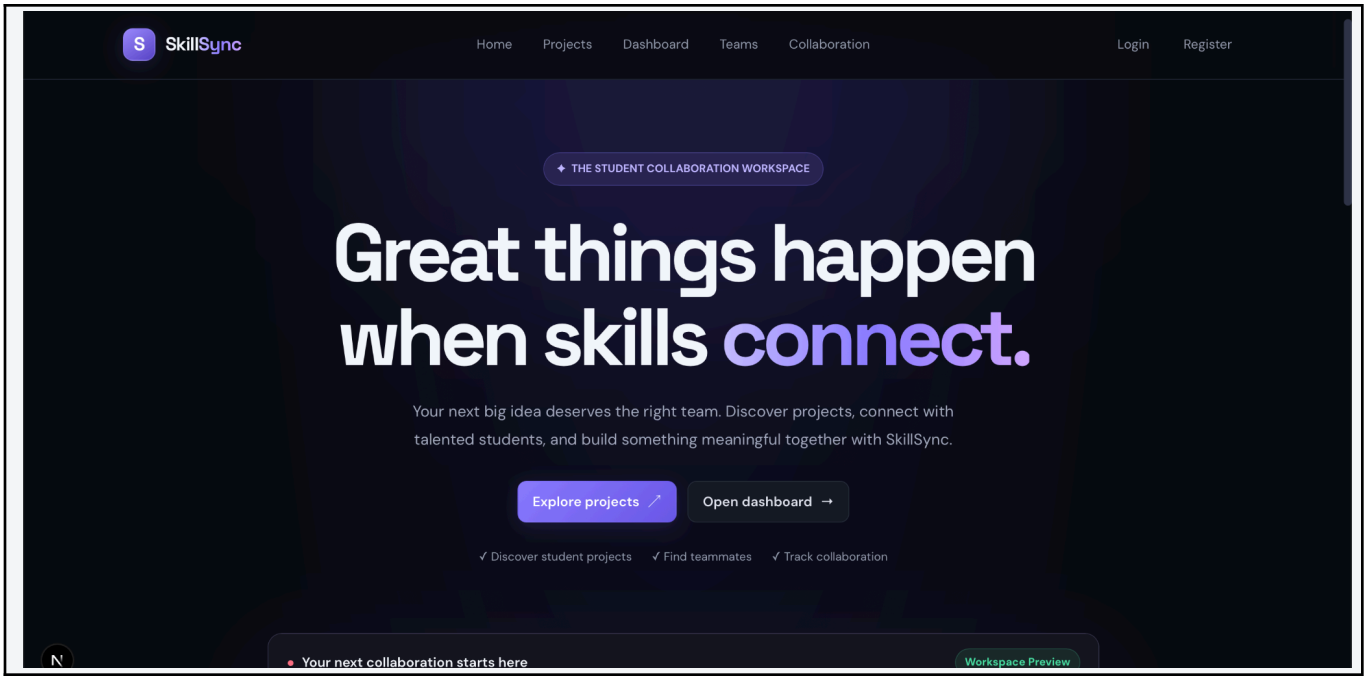
The current implementation uses MongoDB. It does not implement a vector database or vector similarity search.

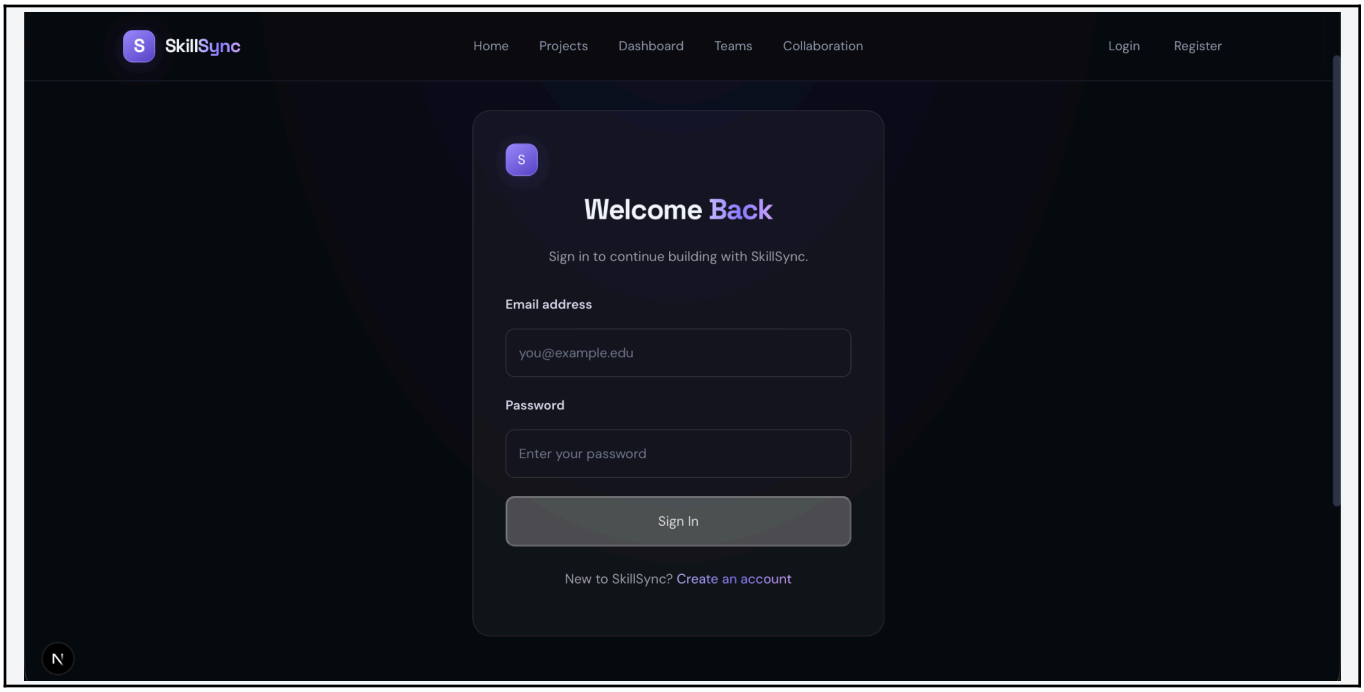
9. Next.js – Project Tree Structure

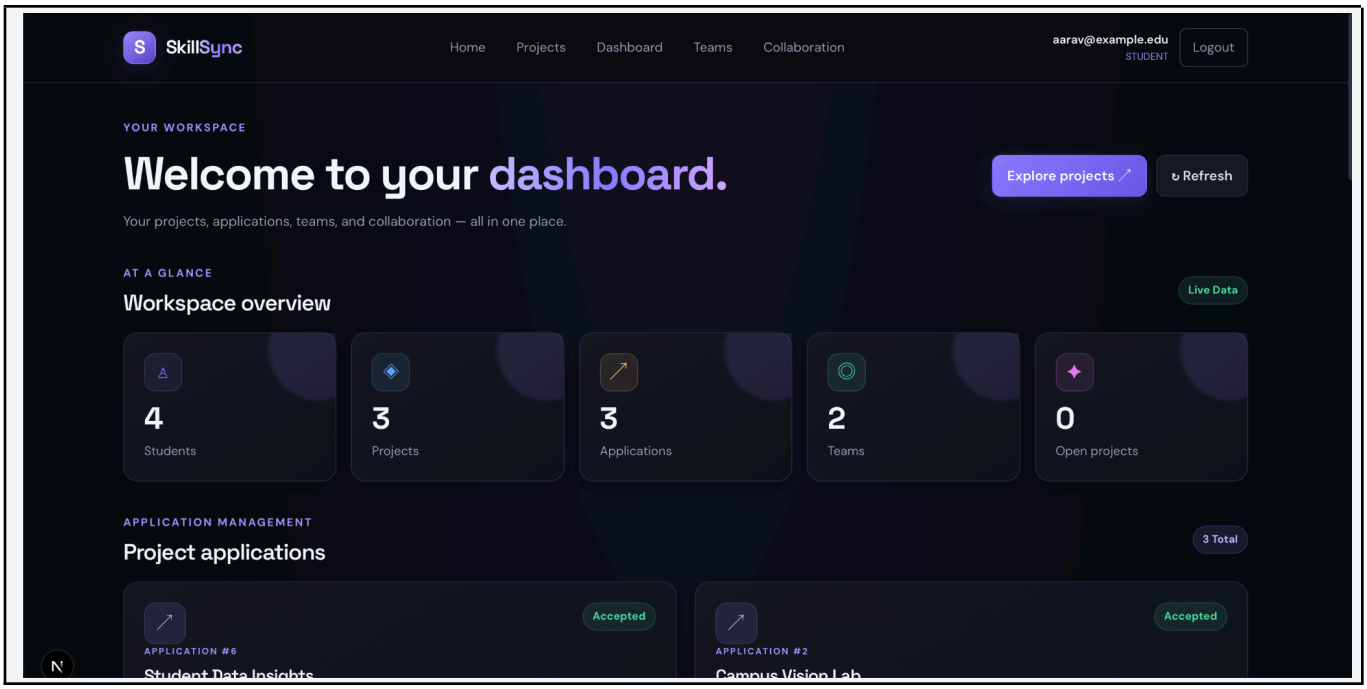
```
SkillSync_Project/\n├── app/\n│   ├── api/\n│   │   ├── applications/accept/route.ts\n│   │   ├── applications/route.ts\n│   │   ├── auth/{login,logout,me,register}/route.ts\n│   │   ├── dashboard/route.ts\n│   │   ├── matches/route.ts\n│   │   ├── mongodb-test/route.ts\n│   │   ├── notes/route.ts\n│   │   ├── projects/route.ts\n│   │   ├── students/route.ts\n│   │   └── teams/{members,route}.ts\n│   ├── collaboration/page.tsx\n│   ├── components/Navbar.tsx\n│   ├── dashboard/page.tsx\n│   ├── login/page.tsx\n│   ├── projects/page.tsx\n│   ├── register/page.tsx\n│   ├── teams/page.tsx\n│   ├── globals.css, layout.tsx, page.tsx\n├── database/{auth-schema,queries,schema,seed}.sql\n├── docs/{architecture,test-plan}.md\n├── lib/{auth-guard,auth,db,mongodb}.ts\n├── .env.example, .gitignore\n└── package.json, package-lock.json, next.config.ts, tsconfig.json
```

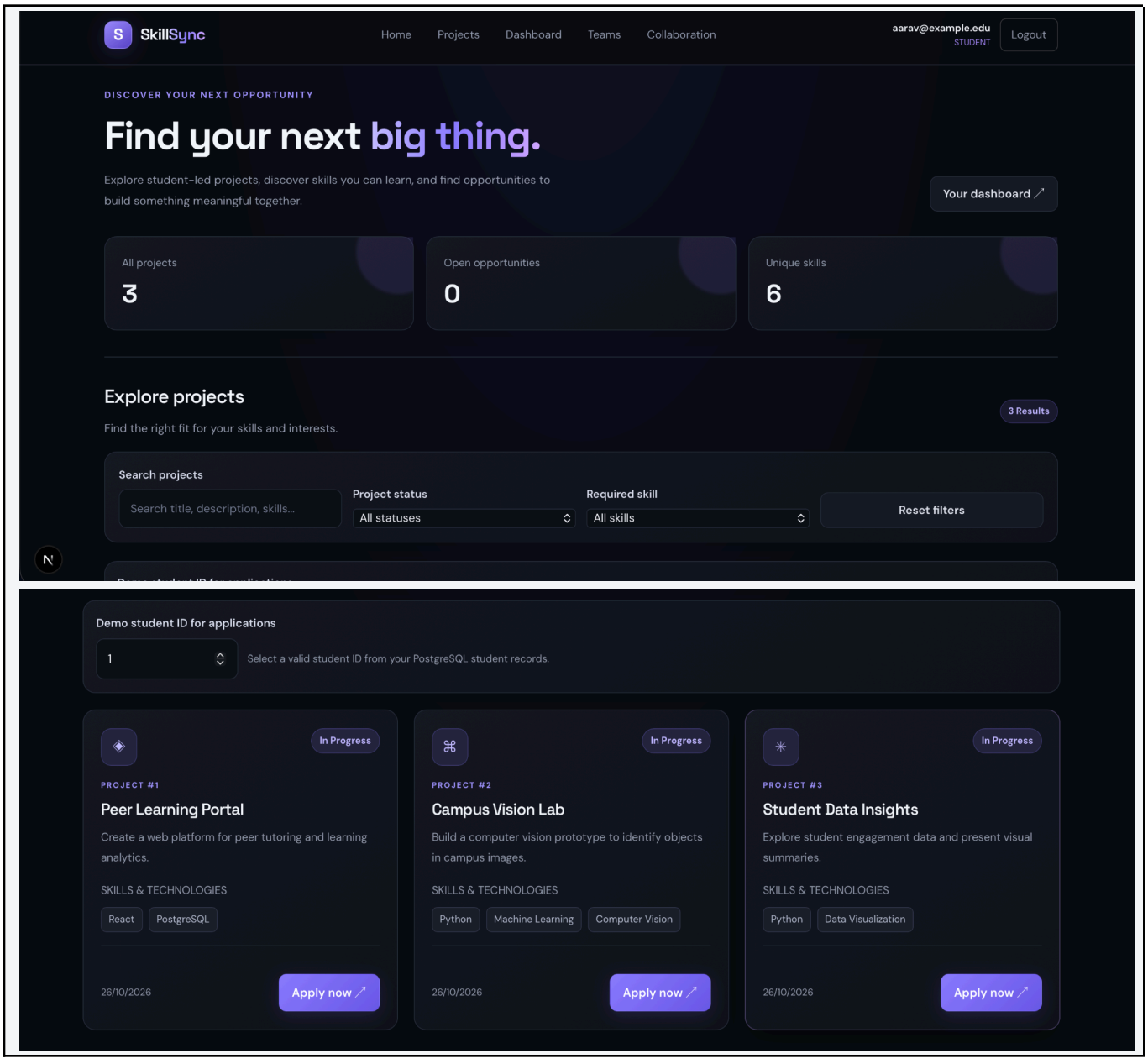
10. Application Screenshots and Demonstration

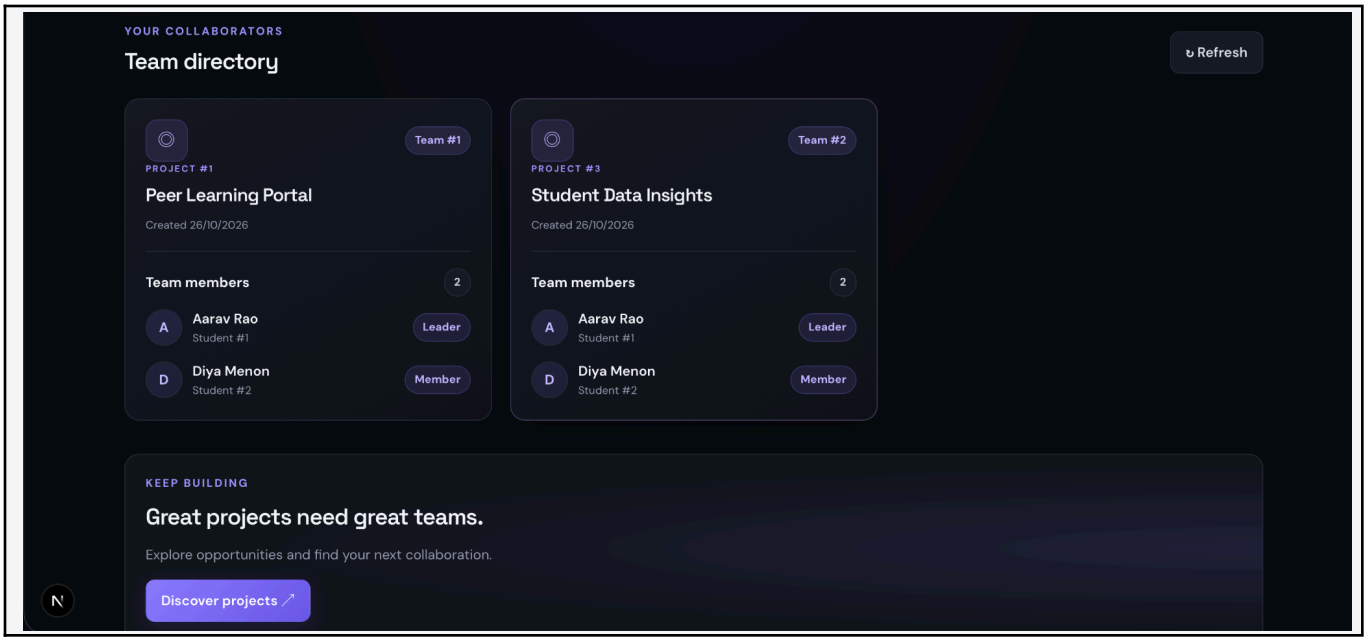
Capture the four application screenshots below from the running application. Use a consistent browser size and make sure the page content is readable. Do not expose passwords, session cookies, or environment secrets.

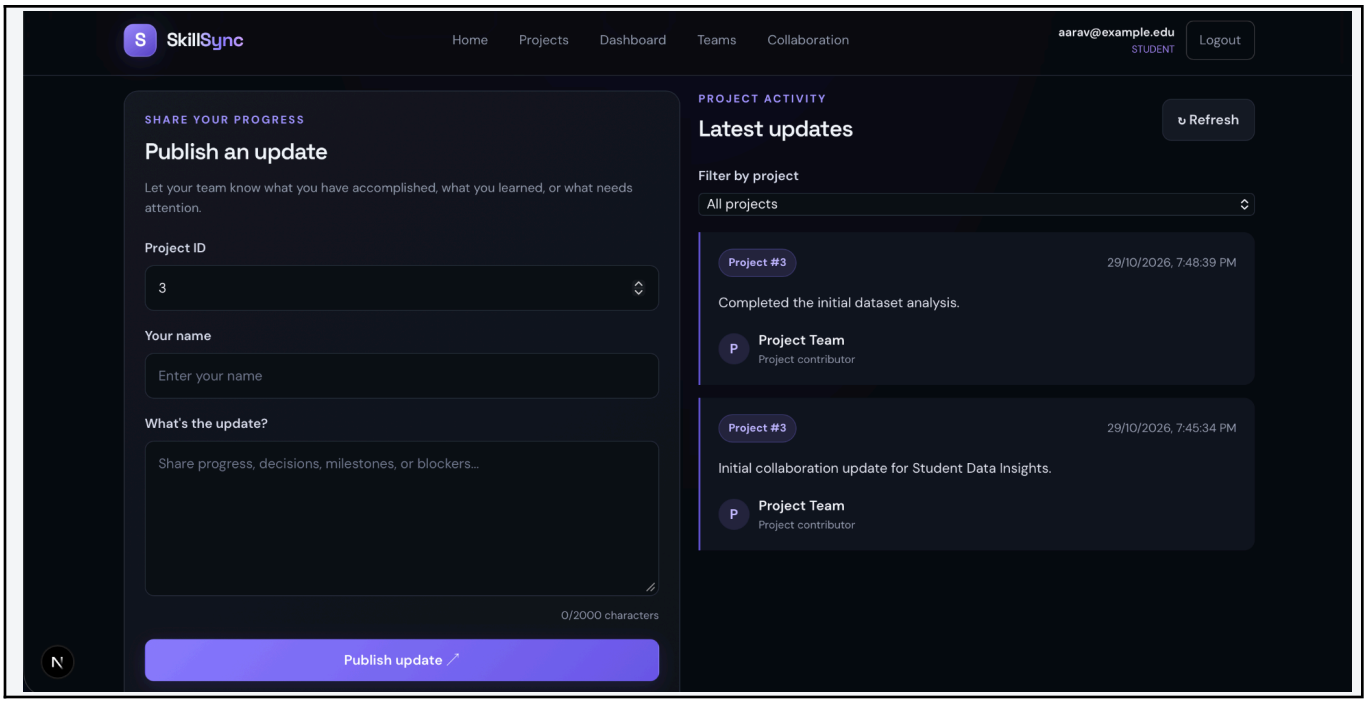



SkillSync
Home
Projects
Dashboard
Teams
Collaboration
Login
Register
Welcome Back
Sign in to continue building with SkillSync.
Email address
you@example.edu
Password
Enter your password
Sign In
New to SkillSync? Create an account


SkillSync
Home
Projects
Dashboard
Teams
Collaboration
aarav@example.edu
STUDENT
Logout
YOUR WORKSPACE
Welcome to your dashboard.
Explore projects
Refresh
Your projects, applications, teams, and collaboration — all in one place.
AT A GLANCE
Workspace overview
Live Data
4
Students
3
Projects
3
Applications
2
Teams
0
Open projects
APPLICATION MANAGEMENT
Project applications
3 Total
Accepted
APPLICATION #6
Accepted
APPLICATION #2


SkillSync
Home
Projects
Dashboard
Teams
Collaboration
aarav@example.edu
STUDENT
Logout
DISCOVER YOUR NEXT OPPORTUNITY
Find your next big thing.
Explore student-led projects, discover skills you can learn, and find opportunities to build something meaningful together.
Your dashboard
All projects
3
Open opportunities
0
Unique skills
6
Explore projects
Find the right fit for your skills and interests.
3 Results
Search projects
Search title, description, skills...
Project status
All statuses
Required skill
All skills
Reset filters
Demo student ID for applications
1
Select a valid student ID from your PostgreSQL student records.
In Progress
PROJECT #1
Peer Learning Portal
Create a web platform for peer tutoring and learning analytics.
SKILLS & TECHNOLOGIES
React
PostgreSQL
26/10/2026
Apply now
In Progress
PROJECT #2
Campus Vision Lab
Build a computer vision prototype to identify objects in campus images.
SKILLS & TECHNOLOGIES
Python
Machine Learning
Computer Vision
26/10/2026
Apply now
In Progress
PROJECT #3
Student Data Insights
Explore student engagement data and present visual summaries.
SKILLS & TECHNOLOGIES
Python
Data Visualization
26/10/2026
Apply now


YOUR COLLABORATORS
Team directory
Refresh
Team #1
PROJECT #1
Peer Learning Portal
Created 26/10/2026
Team members
2
A
Aarav Rao
Student #1
Leader
D
Diya Menon
Student #2
Member
Team #2
PROJECT #3
Student Data Insights
Created 26/10/2026
Team members
2
A
Aarav Rao
Student #1
Leader
D
Diya Menon
Student #2
Member
KEEP BUILDING
Great projects need great teams.
Explore opportunities and find your next collaboration.
Discover projects


SkillSync
Home
Projects
Dashboard
Teams
Collaboration
aarav@example.edu
STUDENT
Logout
SHARE YOUR PROGRESS
Publish an update
Let your team know what you have accomplished, what you learned, or what needs attention.
Project ID
3
Your name
Enter your name
What's the update?
Share progress, decisions, milestones, or blockers...
0/2000 characters
Publish update
PROJECT ACTIVITY
Latest updates
Refresh
Filter by project
All projects
Project #3
29/10/2026, 7:48:39 PM
Completed the initial dataset analysis.
P
Project Team
Project contributor
Project #3
29/10/2026, 7:45:34 PM
Initial collaboration update for Student Data Insights.
P
Project Team
Project contributor

Working:

```
satyakallakuri@Satyas-Laptop SkillSync_Project % psql -d skillsync
psql (16.15 (Homebrew))
Type "help" for help.

skillsync=# \dt
               List of relations
 Schema |     Name      | Type  |     Owner
--------+---------------+-------+----------------
 public | application   | table | satyakallakuri
 public | project       | table | satyakallakuri
 public | project_skill | table | satyakallakuri
 public | skill         | table | satyakallakuri
 public | student       | table | satyakallakuri
 public | student_skill | table | satyakallakuri
 public | team          | table | satyakallakuri
 public | team_member   | table | satyakallakuri
 public | user_account  | table | satyakallakuri
(9 rows)

skillsync=# SELECT * FROM project_summary;
 project_id |         title         |   status    | owner_name | required_skill_count | application_count
------------+-----------------------+-------------+------------+----------------------+-------------------
          1 | Peer Learning Portal  | in_progress | Diya Menon |                    2 |                 1
          2 | Campus Vision Lab     | in_progress | Aarav Rao  |                    3 |                 1
          3 | Student Data Insights | in_progress | Kabir Shah |                    2 |                 1
(3 rows)

(END)
```

```
satyakallakuri@Satyas-Laptop SkillSync_Project % mongosh
Current Mongosh Log ID: 6ae88cbf54fe859bb07b3e0a
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.14.0
Using MongoDB:          9.0.4
Using Mongosh:          2.14.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/


To help improve our products, pseudo-anonymous usage data is collected from interactive and agentic sessions only, and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

------
   The server generated these startup warnings when booting
   2026-10-29T19:41:29.190+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> db.runCommand({ ping: 1 })
{ ok: 1 }
test> use skillsync
| show collections
| db.notes.find().pretty()
switched to db skillsync
skillsync>
```

The four UI screenshots are the main application demonstration set. Figures 5 and 6 provide supporting evidence for the SQL and NoSQL database sections.

11. Evaluation Rubric – Completion Status

Requirement	Status	Evidence / note
Project title and student details	Complete	Sections 1 and 3.
Abstract and technology stack	Complete	Sections 4 and 5.
SQL database commands	Implemented	DDL, DML, joins, aggregates, subqueries, transactions, indexes, view, trigger and full-text search.
NoSQL / vector database commands	Partially fulfilled	MongoDB implemented; vector database/search not implemented.
Next.js project tree	Complete	Section 9.
Application screenshots	Pending insertion	Four UI screenshots to be added in Section 10.
Code/database screenshots	Recommended	Add PostgreSQL and MongoDB evidence in Section 10.
Authentication and authorization	Tested	Student session worked; student was denied admin-only endpoint.
Build and API verification	Tested	Production build succeeded; student APIs returned HTTP 200 in browser testing.
GitHub repository	Complete	https://github.com/satk4ll/DBMS_MP

12. Conclusion and Future Enhancements

SkillSync demonstrates a full-stack student project collaboration application that integrates PostgreSQL and MongoDB. PostgreSQL stores structured student, skill, project, application, and team data. MongoDB stores collaboration notes. The Next.js application provides authenticated API routes and role-aware workflows.

The database implementation includes integrity constraints, indexes, a summary view, a trigger function, full-text search, and transaction-based testing. Authentication uses hashed passwords and signed HttpOnly session cookies, while server-side role checks protect administrative operations.

Future enhancements may include login rate limiting, email verification, audit logging, pagination, more automated integration tests, and an optional vector-search component if semantic matching is required.

Appendix A. GitHub Repository and Setup

GitHub repository: https://github.com/satk4ll/DBMS_MP

The repository includes the Next.js source, API routes, SQL schema and query scripts, MongoDB integration, documentation, and .env.example. The real .env.local file is not committed.

```
# Install dependencies\nnpm install\n\n# Configure local environment variables using .env.example as a guide.\n\n# Start development server\nnpm run dev\n\n# Build for production\nnpm run build
```

PostgreSQL and MongoDB services must be running and local environment variables must be configured before starting the application.